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INTERIM FEDERAL SPECIFICATION

DRUMS, PLASTIC, MOLDED POLYETHYLENE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers molded polyethylene drums. These drums are used for the shipping and storing of liquids, (see 6.1).

1.2 Classification. The polyethylene drums covered by this specification shall be of the following type, grade, and sizes.

1.2.1 Type. Polyethylene drums shall be furnished in only the closed head type.

1.2.2 Grade. Polyethylene drums shall be furnished in only one grade. This grade shall be suitable for food and beverage use, also for general purpose use. The polyethylene material shall be in complete compliance with Food and Drug Administration Regulations governing the raw materials permitted for the fabrication of packaging materials used in the packaging of food and beverage.

1.2.3 Sizes. Polyethylene drums shall be furnished in the following sizes:

15 gallon nominal capacity
30 gallon nominal capacity
55 gallon nominal capacity

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 178 - Shipping Contain Specifications.
21 CFR 121 - Food Additives.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W. Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 638 - Test for Tensile Properties of Plastics.
D 1505 - Test for Density of Plastics by the Density-Gradient Technique.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. Drums shall be made of polyethylene which shall have the following properties:

Property	Specification	ASTM Method
Density range	0.930 - 0.965	D1505
Tensile strength	3,000 p.s.i. min.	D638
Percent elongation	75 percent min.	D638

Ultraviolet light protection shall be provided by impregnation of polyethylene with carbon black or other equally efficient pigments or inhibitors. These additives must be compatible with contents and must retain their effectiveness for the life of the container. These inhibitors or pigments shall not reduce the physical properties of the polyethylene below the required values.

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3.2 Manufacture.

3.2.1 Construction. The drum shall be molded in one piece but may have a multiple wall construction. The dimensions and capacities of drums shall be as specified in Table I.

TABLE I. Construction and dimensions

Gallons	15	30	55
Outside height, inches	24 ± 1	32 ± 2	38 ± 3
Capacity, gallons			
Minimum	15.6	31.2	57.2
Maximum	16.5	33	60.5
Capacity, weight of contents max. lbs.	200	400	750

3.2.2 Closures

3.2.2.1 Flanges. Unless otherwise specified, drums shall have two flanged openings. Drums shall have a 2-inch opening diametrically opposed by 3/4-inch opening. These openings shall be threaded internally to accommodate plastic plugs. Alternately drums may have two 2-inch openings diametrically opposed, one opening containing a 2-inch to 3/4-inch reducing plug.

3.2.2.2 Plugs. Drums shall be provided with plastic plugs for each opening. All plugs, combination or solid, shall be constructed so they may be tightened with a standard plug wrench.

3.3 Leakage. The drum and closures shall not leak when tested as specified in 4.3.

3.4 Identification markings. The bottom of the drum shall be permanently marked to show nominal capacity, manufacturer's name, and date of manufacture.

3.5 Color. The drum shall be furnished in natural and basic colors as specified by the purchaser. Color pigments shall not alter the physical properties of the polyethylene or alter the effectiveness of the ultra violet inhibitors.

3.6 Drum furnished as filled containers. When drums are furnished as filled containers, the drums and closures shall not interact chemically with the chemicals contained therein so as to be themselves altered or so as to alter the strength, quality, or purity of the contents of the containers beyond the requirements specified in the contract or order.

3.7 Workmanship. The drum shall be free from contamination (foreign material) and damage such as cracks, dents, or deformation which would impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all drums submitted for inspection at the same time.

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4.2.2 Visual examination of end item. Sampling for this examination shall be according to level S-3 in MIL-STD-105. The Acceptable Quality Levels (AQL's) for defects listed in table II shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Capacity not as specified	X	
Marking illegible		X
Damage or defects not affecting function or serviceability		X
Damage of defects affecting function or serviceability	X	

4.2.3 Sampling for tests. Sampling for tests shall be in accordance with paragraphs 4.3.1, 4.3.2, and 4.3.3.

4.3 Tests.

4.3.1 Tests for drums. At least three samples taken at random, filled and prepared as specified and closed as for use, shall be capable of withstanding the tests in paragraphs 4.3.1.1, 4.3.1.2, and 4.3.1.3 without leaking. These tests shall be performed at the start of initial production and at 4-month intervals and shall be repeated on any change of type, size, materials, or process method. No single container shall be expected to withstand more than one of the following tests:

4.3.1.1 The container filled to 98 percent capacity with water shall be dropped from a height of 4 feet onto solid concrete so as to drop diagonally on top edge or any part constructed to a lesser strength.

4.3.1.2 The container filled 98 percent capacity with a solution compatible with polyethylene and which remains liquid at 0 degrees F. shall be dropped from a height of 4 feet onto solid concrete on any part of the container when container and contents are at or slightly below 0 degrees F. Filled container shall be stored at 0 degrees F. or lower temperature for at least 4 hours immediately preceding test.

4.3.1.3 The container shall be tested by retaining for 5 minutes hydrostatic pressure of at least 15 pounds per square inch at equilibrium without showing pressure drop or evidence of leakage.

4.3.2 At least three containers taken at random from each continuous production lot of more than 1,000 containers of each given type and size shall withstand without leakage or failure the test prescribed in 4.3.1.2.

4.3.3 At least three containers of each size and type taken at random at start of initial production and upon any change in materials, design, or process method shall withstand without failure or leakage the following tests. No single container shall be expected to withstand more than one test:

4.3.3.1 The container filled to 98 percent of capacity with water shall be capable of withstanding a vibration test by placing the container on the vibration table anchored in such manner that all horizontal motion shall be restricted and only vertical motion allowed. The test shall be performed for 1 hour using an amplitude of 1 inch at a frequency that causes the test container to be raised from the floor of the table to such a degree than a piece of paper or flat steel strap or tape can be passed between the table and the container.

4.3.3.2 The container filled to 98 percent capacity with water shall withstand the following static compression test without buckling of the side walls sufficient to cause damage, but in no case shall the maximum top to bottom deflection be more than 1 inch. Compression shall be applied to the load bearing areas of the top of the container for a period of not less than 48 hours.

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Marked (rated) capacity (gallons)	Compression test (pounds)
15	1200
30	1800
55	2400

5. PREPARATION FOR DELIVERY

5.1 Packing. The drums shall be prepared for shipment in a manner to insure carrier acceptance and safe delivery at destination. Shipment shall be in accordance with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable

5.2.1 Civil agencies. All shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. All shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Polyethylene drums covered by this specification are intended for use as containers for liquids. They are not intended for use with strong oxidizing liquids or aromatic or aliphatic hydrocarbons, but are suitable for some organic and inorganic acids, alcohols, and alkalies. In all cases, the chemical should not be packaged in the polyethylene drum if the boiling point of the chemical approximates or is lower than the anticipated storage temperature.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size (see 1.2).
(c) Color (see 3.5).
(d) Size of lot, if different from 4.2.1.
(e) Type and size of closures, if other than standard (see 3.2.2).
(f) Special marking when required.

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